

Guided Image Filtering

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Abstract. In this paper, we propose a novel type of explicit image filter - *guided filter*. Derived from a local linear model, the guided filter generates the filtering output by considering the content of a guidance image, which can be the input image itself or another different image. The guided filter can perform as an edge-preserving smoothing operator like the popular bilateral filter [1], but has better behavior near the edges. It also has a theoretical connection with the matting Laplacian matrix [2], so is a more generic concept than a smoothing operator and can better utilize the structures in the guidance image. Moreover, the guided filter has a fast and non-approximate linear-time algorithm, whose computational complexity is independent of the filtering kernel size. We demonstrate that the guided filter is both effective and efficient in a great variety of computer vision and computer graphics applications including noise reduction, detail smoothing/enhancement, HDR compression, image matting/feathering, haze removal, and joint upsampling.

1 Introduction

Most applications in computer vision and computer graphics involve the concept of image filtering to reduce noise and/or extract useful image structures. Simple explicit linear translation-invariant (LTI) filters like Gaussian filter, Laplacian filter, and Sobel filter are widely used in image blurring/sharpening, edge detection, and feature extraction [3]. LTI filtering also includes the process of solving a Poisson Equation, such as in high dynamic range (HDR) compression [4], image stitching [5], and image matting [6], where the filtering kernel is implicitly defined by the inverse of a homogenous Laplacian matrix.

The kernels of LTI filters are spatially invariant and independent of any image content. But in many cases, we may want to incorporate additional information from a given *guidance* image during the filtering process. For example, in colorization [7] the output chrominance channels should have consistent edges with the given luminance channel; in image matting [2] the output alpha matte should capture the thin structures like hair in the image. One approach to achieve this purpose is to optimize a quadratic function that directly enforces some constraints on the unknown output by considering the guidance image. The solution is then obtained by solving a large sparse matrix encoded with the information of the guidance image. This inhomogeneous matrix implicitly defines a *translation-variant* filtering kernel. This approach is widely used in many

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Abstract. 在本文中, 我们提出了一种新型的显式图像滤波器——*guided filter*。该滤波器基于局部线性模型, 通过考虑引导图像的内容生成滤波输出, 引导图可以是输入图像本身或其他不同的图像。导向滤波器可作为边缘保持平滑算子使用, 类似于流行的双边滤波器[1], 但在边缘附近表现更好。它还与蒙版拉普拉斯矩阵[2]有理论联系, 因此比单纯的平滑算子更为通用, 并能更好地利用引导图中的结构信息。此外, 导向滤波器具有快速且非近似的线性时间算法, 其计算复杂度与滤波核大小无关。我们展示了该导向滤波器在噪声减少、细节平滑/增强、HDR压缩、图像蒙版/羽化、雾霾去除和联合上采样等多种计算机视觉和计算机图形学应用中均有效且高效。

1 Introduction

大多数计算机视觉和计算机图形学应用都涉及图像滤波的概念, 用于减少噪声或提取有用的图像结构。简单的显式线性平移不变 (LTI) 滤波器, 如高斯滤波器、拉普拉斯滤波器和索贝尔滤波器, 在图像模糊/锐化、边缘检测和特征提取中被广泛使用[3]。LTI滤波还包括求解泊松方程的过程, 例如在高动态范围 (HDR) 压缩[4]、图像拼接[5]和图像抠图[6]等场景中, 其中滤波核由拉普拉斯矩阵的逆矩阵隐式定义。

LTI滤波器的核在空间上是不变的, 并且与任何图像内容无关。但在许多情况下, 我们可能希望在滤波过程中从给定的 *guidance* 图像中获取额外的信息。例如, 在色彩化[7]中, 输出的颜色通道应与给定的亮度通道具有一致的边缘; 在图像抠图[2]中, 输出的alpha蒙版应该捕捉到头发这样的细小结构。一种实现这一目标的方法是优化一个二次函数, 该函数直接通过考虑引导图像来对未知输出施加某些约束条件。然后通过求解一个包含引导图像信息的大稀疏矩阵得到解决方案。这个非齐次矩阵隐式地定义了一个 *translation-variant* 滤波核。这种方法在许多应用场景中被广泛使用。

applications, like colorization [7], image matting [2], multi-scale decomposition [8], and haze removal [9]. While this optimization-based approach often yields the state-of-the-art quality, it comes with the price of long computational time.

The other approach is to explicitly build the filter kernels using the guidance image. The bilateral filter, proposed in [10], made popular in [1], and later generalized in [11], is perhaps the most popular one of such filters. Its output at a pixel is a weighted average of the nearby pixels, where the weights depend on the intensity/color similarities in the guidance image. The guidance image can be the filter input itself [1] or another image [11]. The bilateral filter can smooth small fluctuations and preserve edges. While this filter is effective in many situations, it may have unwanted gradient reversal artifacts [12, 13, 8] near edges (further explained in Section 3.4). Its fast implementation is also a challenging problem. Recent techniques [14–17] rely on quantization methods to accelerate but may sacrifice the accuracy.

In this paper we propose a new type of explicit image filter, called *guided filter*. The filtering output is locally a linear transform of the guidance image. This filter has the edge-preserving smoothing property like the bilateral filter, but does not suffer from the gradient reversal artifacts. It is also related to the matting Laplacian matrix [2], so is a more generic concept and is applicable in other applications beyond the scope of "smoothing". Moreover, the guided filter has an $O(N)$ time (in the number of pixels N) *exact* algorithm for both gray-scale and color images. Experiments show that the guided filter performs very well in terms of both quality and efficiency in a great variety of applications, such as noise reduction, detail smoothing/enhancement, HDR compression, image matting/feathering, haze removal, and joint upsampling.

2 Related Work

2.1 Bilateral Filter

The bilateral filter computes the filter output at a pixel as a weighted average of neighboring pixels. It smooths the image while preserving edges. Due to this nice property, it has been widely used in noise reduction [18], HDR compression [12], multi-scale detail decomposition [19], and image abstraction [20]. It is generalized to the joint bilateral filter in [11], in which the weights are computed from another guidance image rather than the filter input. The joint bilateral filter is particular favored when the filter input is not reliable to provide edge information, e.g., when it is very noisy or is an intermediate result. The joint bilateral filter is applicable in flash/no-flash denoising [11], image upsampling [21], and image deconvolution [22].

However, it has been noticed [12, 13, 8] that the bilateral filter may have the gradient reversal artifacts in detail decomposition and HDR compression. The reason is that when a pixel (often on an edge) has few similar pixels around it, the Gaussian weighted average is unstable. Another issue concerning the bilateral filter is its efficiency. The brute-force implementation is in $O(Nr^2)$

应用程序，如着色[7]、图像抠图[2]、多尺度分解[8]和雾气去除[9]。虽然这种基于优化的方法通常能获得最先进的质量，但也伴随着较长的计算时间代价。

另一种方法是利用引导图像显式构建滤波核。双边滤波器，最早在[10]中提出，在[1]中流行起来，并在[11]中进一步推广，可能是此类滤波器中最受欢迎的一种。该滤波器在一个像素上的输出是附近像素的加权平均值，其中权重取决于引导图像中的强度/颜色相似性。引导图可以是滤波输入本身[1]或另一幅图像[11]。双边滤波器能够平滑小波动并保留边缘。虽然这种滤波器在许多情况下都很有效，但在边缘附近可能会产生不希望的梯度反转伪影[12, 13, 8]（详见第3.4节）。其快速实现也是一个具有挑战性的问题。最近的技术[14–17]依赖于量化方法来加速处理速度，但可能牺牲准确性。

在本文中，我们提出了一种新的显式图像滤波器，称为*guided filter*。该滤波器的输出是指导图像的局部线性变换结果。此滤波器具有双边滤波器保持边缘平滑的特性，但不会产生梯度反转伪影。它还与蒙版拉普拉斯矩阵[2]相关联，因此是一个更通用的概念，并且在“平滑”之外的应用中也适用。此外，引导滤波器对于灰度和彩色图像都具有时间复杂度为 $O(N)$ （像素数量 N ）的算法 *exact*。实验表明，在噪声减少、细节平滑/增强、HDR压缩、图像蒙版/羽化、雾霾去除以及联合上采样等多种应用中，引导滤波器在质量和效率方面均表现出色。

2 Related Work

2.1 Bilateral Filter

双边滤波器在计算像素的滤波输出时，会将其作为相邻像素加权平均的结果。它可以在平滑图像的同时保留边缘信息。由于这一优良特性，双边滤波器被广泛应用于噪声减少[18]、HDR压缩[12]、多尺度细节分解[19]和图像抽象化[20]等领域。在文献[11]中，双边滤波器被推广为联合双边滤波器，在这种情况下，权重是根据另一幅引导图计算得出的，而不是直接基于滤波输入。当滤波输入无法可靠地提供边缘信息时（例如，噪声非常大或结果为中间值），联合双边滤波器特别受到青睐。联合双边滤波器适用于闪光/无闪光去噪[11]、图像上采样[21]和图像反卷积[22]等场景。

然而，已有研究指出[12, 13, 8]，双边滤波在细节分解和HDR压缩中可能会出现梯度反转的伪影。原因是当一个像素（通常位于边缘）周围缺乏相似像素时，高斯加权平均值会变得不稳定。另一个问题是双边滤波的效率问题。暴力实现方法的时间复杂度为 $O(v^6)$ 。

time, which is prohibitively high when the kernel radius r is large. In [14] an approximated solution is obtained in a discretized space-color grid. Recently, $O(N)$ time algorithms [15, 16] have been developed based on histograms. Adams et al. [17] propose a fast algorithm for color images. All the above methods require a high quantization degree to achieve satisfactory speed, but at the expense of quality degradation.

2.2 Optimization-based Image Filtering

A series of approaches optimize a quadratic cost function and solve a linear system, which is equivalent to implicitly filtering an image by an inverse matrix. In image segmentation [23] and colorization [7], the affinities of this matrix are Gaussian functions of the color similarities. In image matting, a matting Laplacian matrix [2] is designed to enforce the alpha matte as a local linear transform of the image colors. This matrix is also applicable in haze removal [9]. The weighted least squares (WLS) filter in [8] adjusts the matrix affinities according to the image gradients and produces a halo-free decomposition of the input image. Although these optimization-based approaches often generate high quality results, solving the corresponding linear system is time-consuming.

It has been found that the optimization-based filters are closely related to the explicit filters. In [24] Elad shows that the bilateral filter is one Jacobi iteration in solving the Gaussian affinity matrix. In [25] Fattal defines the edge-avoiding wavelets to approximate the WLS filter. These explicit filters are often simpler and faster than the optimization-based filters.

3 Guided Filter

We first define a general linear translation-variant filtering process, which involves a guidance image I , an input image p , and an output image q . Both I and p are given beforehand according to the application, and they can be identical. The filtering output at a pixel i is expressed as a weighted average:

$$q_i = \sum_j W_{ij}(I) p_j, \quad (1)$$

where i and j are pixel indexes. The filter kernel W_{ij} is a function of the guidance image I and independent of p . This filter is linear with respect to p .

A concrete example of such a filter is the joint bilateral filter [11]. The bilateral filtering kernel W^{bf} is given by:

$$W_{ij}^{\text{bf}}(I) = \frac{1}{K_i} \exp\left(-\frac{|\mathbf{x}_i - \mathbf{x}_j|^2}{\sigma_s^2}\right) \exp\left(-\frac{|I_i - I_j|^2}{\sigma_r^2}\right). \quad (2)$$

where $\mathbf{x}$ is the pixel coordinate, and K_i is a normalizing parameter to ensure that $\sum_j W_{ij}^{\text{bf}} = 1$. The parameters σ_s and σ_r adjust the spatial similarity and the range (intensity/color) similarity respectively. The joint bilateral filter degrades to the original bilateral filter [1] when I and p are identical.

当内核半径 r 较大时，所需时间非常高。文献 [14] 中提出了一种在离散的空间-颜色网格中获得的近似解。最近，基于直方图的方法开发出了 $O(N)$ 时间复杂度的算法 [15, 16]。Adams 等人 [17] 提出了一种针对彩色图像的快速算法。以上所有方法都需要较高的量化程度以达到满意的运行速度，但这也导致了质量下降。

2.2 Optimization-based Image Filtering

一系列方法优化二次成本函数并求解线性系统，这相当于隐式地使用逆矩阵对图像进行滤波。在图像分割[23]和着色[7]中，该矩阵的亲合力是颜色相似度的高斯函数。在图像抠图中，设计了一种抠图拉普拉斯矩阵[2]，以强制 α 蒙版作为图像颜色的局部线性变换。此矩阵同样适用于去除雾霾[9]。文献[8]中的加权最小二乘(WLS)滤波器根据图像梯度调整矩阵亲和力，并生成无光晕效应的输入图像分解。尽管这些基于优化的方法通常会产生高质量的结果，但求解相应的线性系统耗时较长。

基于优化的滤波器与显式滤波器密切相关。在文献[24]中，Elad 表明双边滤波器是求解高斯相似矩阵的一个 Jacobi 迭代步骤。Fattal 在文献[25]中定义了边缘规避小波来近似 WLS 滤波器。这些显式滤波器通常比基于优化的滤波器更简单、更快。

3 Guided Filter

我们首先定义一个一般的线性时变滤波过程，该过程涉及引导图像 I 、输入图像 p 和输出图像 q 。根据具体应用， I 和 p 事先给定，并且它们可以相同。像素 i 处的滤波输出表示为加权平均值：

$$q_i = \sum_j W_{ij}(I) p_j, \quad (1)$$

其中 i 和 j 是像素索引。滤波核 W_{ij} 是指导图像 I 的函数，并且独立于 p 。该滤波器相对于 p 是线性的。

一个具体的滤波器示例是联合双边滤波器[11]。双边滤波核 W^{bf} 由以下公式给出：

$$W_{ij}^{\text{bf}}(I) = \frac{1}{K_i} \exp\left(-\frac{|\mathbf{x}_i - \mathbf{x}_j|^2}{\sigma_s^2}\right) \exp\left(-\frac{|I_i - I_j|^2}{\sigma_r^2}\right). \quad (2)$$

其中， $\mathbf{x}$ 是像素坐标， K_i 是一个归一化参数，以确保 $\sum_j W_{ij}^{\text{bf}} = 1$ 。参数 σ_s 和 σ_r 分别调整空间相似性和范围（强度/颜色）相似性。当 I 和 p 相同时，联合双边滤波器退化为原始的双边滤波器 [1]。

3.1 Definition

Now we define the guided filter and its kernel. The key assumption of the guided filter is a local linear model between the guidance I and the filter output q . We assume that q is a linear transform of I in a window ω_k centered at the pixel k :

$$q_i = a_k I_i + b_k, \forall i \in \omega_k, \quad (3)$$

where (a_k, b_k) are some linear coefficients assumed to be constant in ω_k . We use a square window of a radius r . This local linear model ensures that q has an edge only if I has an edge, because $\nabla q = a \nabla I$. This model has been proven useful in image matting [2], image super-resolution [26], and haze removal [9].

To determine the linear coefficients, we seek a solution to (3) that minimizes the difference between q and the filter input p . Specifically, we minimize the following cost function in the window:

$$E(a_k, b_k) = \sum_{i \in \omega_k} ((a_k I_i + b_k - p_i)^2 + \epsilon a_k^2). \quad (4)$$

Here ϵ is a regularization parameter preventing a_k from being too large. We will investigate its significance in Section 3.2. The solution to (4) can be given by linear regression [27]:

$$a_k = \frac{\frac{1}{|\omega|} \sum_{i \in \omega_k} I_i p_i - \mu_k \bar{p}_k}{\sigma_k^2 + \epsilon} \quad (5)$$

$$b_k = \bar{p}_k - a_k \mu_k. \quad (6)$$

Here, μ_k and σ_k^2 are the mean and variance of I in ω_k , $|\omega|$ is the number of pixels in ω_k , and $\bar{p}_k = \frac{1}{|\omega|} \sum_{i \in \omega_k} p_i$ is the mean of p in ω_k .

Next we apply the linear model to all local windows in the entire image. However, a pixel i is involved in all the windows ω_k that contain i , so the value of q_i in (3) is not the same when it is computed in different windows. A simple strategy is to average all the possible values of q_i . So after computing (a_k, b_k) for all patches ω_k in the image, we compute the filter output by:

$$q_i = \frac{1}{|\omega|} \sum_{k: i \in \omega_k} (a_k I_i + b_k) \quad (7)$$

$$= \bar{a}_i I_i + \bar{b}_i \quad (8)$$

where $\bar{a}_i = \frac{1}{|\omega|} \sum_{k \in \omega_i} a_k$ and $\bar{b}_i = \frac{1}{|\omega|} \sum_{k \in \omega_i} b_k$.

With this modification ∇q is no longer scaling of ∇I , because the linear coefficients $(\bar{a}_i, \bar{b}_i)$ vary spatially. But since $(\bar{a}_i, \bar{b}_i)$ are the output of an average filter, their gradients should be much smaller than that of I near strong edges. In this situation we can still have $\nabla q \approx \bar{a} \nabla I$, meaning that abrupt intensity changes in I can be mostly maintained in q .

We point out that the relationship among I , p , and q given by (5), (6), and (8) are indeed in the form of image filtering (1). In fact, a_k in (5) can be

3.1 Definition

现在我们定义导向滤波器及其核函数。导向滤波器的关键假设是在以像素 k 为中心的窗口 ω_k 内，指导信号 I 和滤波输出 q 之间存在局部线性模型。我们认为 q 是 I 在该窗口内的线性变换：

$$q_i = a_k I_i + b_k, \forall i \in \omega_k, \quad (3)$$

其中 (a_k, b_k) 是一些假设在 ω_k 中为常数的线性系数。我们使用一个半径为 r 的方形窗口。这种局部线性模型确保了当且仅当 I 存在边缘时， q 才会有边缘，因为 $\nabla q = a \nabla I$ 。该模型已在图像抠图 [2]、超分辨率图像重建 [26] 以及去除雾霾 [9] 等领域证明是有效的。

为了确定线性系数，我们寻求一种解决方案，使窗口内的 q 与滤波器输入 p 之间的差异最小化。具体而言，我们最小化以下成本函数：

$$E(a_k, b_k) = \sum_{i \in \omega_k} ((a_k I_i + b_k - p_i)^2 + \epsilon a_k^2). \quad (4)$$

这里 ϵ 是一个正则化参数，用于防止 a_k 过大。我们将在第3.2节中探讨其重要性。方程 (4) 的解可以通过线性回归[27]给出：

$$a_k = \frac{\frac{1}{|\omega|} \sum_{i \in \omega_k} I_i p_i - \mu_k \bar{p}_k}{\sigma_k^2 + \epsilon} \quad (5)$$

$$b_k = \bar{p}_k - a_k \mu_k. \quad (6)$$

在这里， μ_k 和 σ_k^2 分别是 ω_k 中 I 的均值和方差， $|\omega|$ 是 ω_k 中的像素数量，而 $\bar{p}_k = \frac{1}{|\omega|} \sum_{i \in \omega_k} p_i$ 是 ω_k 中 p 的均值。

接下来，我们将线性模型应用于整个图像中的所有局部窗口。然而，像素 i 涉及到所有包含 i 的窗口 ω_k 中，因此在不同窗口中计算 (3) 式的值 q_i 并不相同。一个简单的策略是对所有可能的值 q_i 进行平均。所以在为图像中的所有补丁 ω_k 计算 (a_k, b_k) 后，我们通过以下方式计算滤波器输出：

$$q_i = \frac{1}{|\omega|} \sum_{k: i \in \omega_k} (a_k I_i + b_k) \quad (7)$$

$$= \bar{a}_i I_i + \bar{b}_i \quad (8)$$

其中 $\bar{a}_i = \frac{1}{|\omega|} \sum_{k \in \omega_i} a_k$ 和 $\bar{b}_i = \frac{1}{|\omega|} \sum_{k \in \omega_i} b_k$ 。

经过此修改后， ∇q 不再与 ∇I 成比例，因为线性系数 $(\bar{a}_i, \bar{b}_i)$ 在空间上是变化的。但由于 $(\bar{a}_i, \bar{b}_i)$ 是一个平均滤波器的输出，它们的梯度在强边缘附近应远小于 I 的梯度。在这种情况下，我们仍然可以保持 $\nabla q \approx \bar{a} \nabla I$ ，这意味着 I 中的突然强度变化可以在很大程度上保留在 q 中。

我们指出，由方程 (5)、(6) 和 (8) 给出的 I 、 p 和 q 之间的关系确实符合图像滤波的形式 (1)。实际上，方程 (5) 中的 a_k 可以被

rewritten as a weighted sum of p : $a_k = \sum_j A_{kj}(I)p_j$, where A_{ij} are the weights only dependent on I . For the same reason, we also have $b_k = \sum_j B_{kj}(I)p_j$ from (6) and $q_i = \sum_j W_{ij}(I)p_j$ from (8). It can be proven (see the supplementary materials) that the kernel weights can be explicitly expressed by:

$$W_{ij}(I) = \frac{1}{|\omega|^2} \sum_{k:(i,j) \in \omega_k} \left(1 + \frac{(I_i - \mu_k)(I_j - \mu_k)}{\sigma_k^2 + \epsilon}\right). \quad (9)$$

Some further computations show that $\sum_j W_{ij}(I) = 1$. No extra effort is needed to normalize the weights.

3.2 Edge-preserving Filtering

Fig. 1 (top) shows an example of the guided filter with various sets of parameters. We can see that it has the edge-preserving smoothing property. This can be explained intuitively as following. Consider the case that $I = p$. It is clear that if $\epsilon = 0$, then the solution to (4) is $a_k = 1$ and $b_k = 0$. If $\epsilon > 0$, we can consider two cases:

Case 1: "Flat patch". If the image I is constant in ω_k , then (4) is solved by $a_k = 0$ and $b_k = \bar{p}_k$;

Case 2: "High variance". If the image I changes a lot within ω_k , then a_k becomes close to 1 while b_k is close to 0.

When a_k and b_k are averaged to get $\bar{a}_i$ and $\bar{b}_i$, combined in (8) to get the output, we have that if a pixel is in the middle of a "high variance" area, then its value is unchanged, whereas if it is in the middle of a "flat patch" area, its value becomes the average of the pixels nearby.

More specifically, the criterion of a "flat patch" or a "high variance" is given by the parameter ϵ . The patches with variance (σ^2) much smaller than ϵ are smoothed, whereas those with variance much larger than ϵ are preserved. The effect of ϵ in the guided filter is similar with the range variance σ_r^2 in the bilateral filter (2). Both parameters determine "what is an edge/a high variance patch that should be preserved". Fig. 1 (bottom) shows the bilateral filter results as a comparison.

3.3 Filter Kernel

The edge-preserving smoothing property can also be understood by investigating the filter kernel (9). Take an ideal step edge of a 1-D signal as an example (Fig. 2). The terms $I_i - \mu_k$ and $I_j - \mu_k$ have the same sign (+/-) when I_i and I_j are on the same side of an edge, while they have opposite signs when the two pixels are on different sides. So in (9) the term $1 + \frac{(I_i - \mu_k)(I_j - \mu_k)}{\sigma_k^2 + \epsilon}$ is much smaller (and close to zero) for two pixels on different sides than on the same sides. This means that the pixels across an edge are almost not averaged together. We can also understand the smoothing effect of ϵ from (9). When $\sigma_k^2 \ll \epsilon$ ("flat patch"), the kernel becomes $W_{ij}(I) = \frac{1}{|\omega|^2} \sum_{k:(i,j) \in \omega_k} 1$: this is a low-pass filter that biases neither side of an edge.

重写为 p 的加权和: $a_k = \sum_j A_{kj}(I)p_j$, 其中 A_{ij} 是仅依赖于 I 的权重。出于同样的原因, 我们还可以从 (6) 式得到 $b_k = \sum_j B_{kj}(I)p_j$, 从 (8) 式得到 $q_i = \sum_j W_{ij}(I)p_j$ 。可以通过查阅补充材料证明, 核权重可以明确表示为:

$$W_{ij}(I) = \frac{1}{|\omega|^2} \sum_{k:(i,j) \in \omega_k} \left(1 + \frac{(I_i - \mu_k)(I_j - \mu_k)}{\sigma_k^2 + \epsilon}\right). \quad (9)$$

进一步的计算表明, 对于 $\sum_j W_{ij}(I) = \text{版本}$, 无需额外努力来规范化权重。

3.2 Edge-preserving Filtering

图1 (上部) 展示了具有不同参数集的引导滤波器的一个例子。可以看出, 它具有边缘保持平滑的特性。直观解释如下: 考虑情况 $I = p$ 。显然, 如果 $\epsilon = 0$, 则方程(4)的解是 $a_k = 1$ 且 $b_k = 0$ 。若 $\epsilon > 0$ 不为0, 我们可以考虑两种情形:

情形1: “平坦补丁”。如果图像在 ω_k 上是常数, 则 (4) 由 $a_k = 0$ 和 $b_k = \bar{p}_k$ 解。

情况2: “高方差”。如果图像在 ω_k 内变化很大, 则 a_k 接近于1而 b_k 接近于0。

当计算 a_k 和 b_k 的平均值以得到 $\bar{a}_i$ 和 $\bar{b}_i$, 并将它们结合到公式 (8) 中以获得输出时, 如果一个像素位于“高方差”区域的中心, 则其值保持不变; 而如果该像素位于“平坦区域”的中心, 则其值变为附近像素的平均值。

更具体地说, “平坦补丁”或“高方差”的标准由参数 ϵ 给出。方差 (σ^2) 远小于 ϵ 的补丁会被平滑处理, 而方差远大于 ϵ 的补丁则被保留。在引导滤波器中, ϵ 的作用与双边滤波器 (2) 中的范围方差 σ_r^2 类似。这两个参数共同决定了“什么是应该被保留的边缘/高方差补丁”。图1 (底部) 展示了双边滤波器的结果以作比较。

3.3 Filter Kernel

边缘保持平滑特性也可以通过研究滤波器核 (9) 来理解。以一维信号的理想阶梯边缘为例 (图2)。当 I_i 和 I_j 位于同一侧时, 项集 $I_i - \mu_k$ 和 $I_j - \mu_k$ 具有相同的符号 (+/-), 而当这两个像素位于不同侧时, 则符号相反。因此, 在 (9) 中, 对于位于不同侧的两个像素而言, 项 $1 + \frac{(I_i - \mu_k)(I_j - \mu_k)}{\sigma_k^2 + \epsilon}$ 要小得多 (接近于零)。这意味着跨边缘的像素几乎不会被平均在一起。我们也可以从 (9) 理解 ϵ 的平滑效果。当 $\sigma_k^2 \ll \epsilon$ (“平坦区域”) 时, 核变为 $W_{ij}(I) = \frac{1}{|\omega|^2} \sum_{k:(i,j) \in \omega_k} 1$: 这是一个低通滤波器, 对边缘两侧没有偏向性。

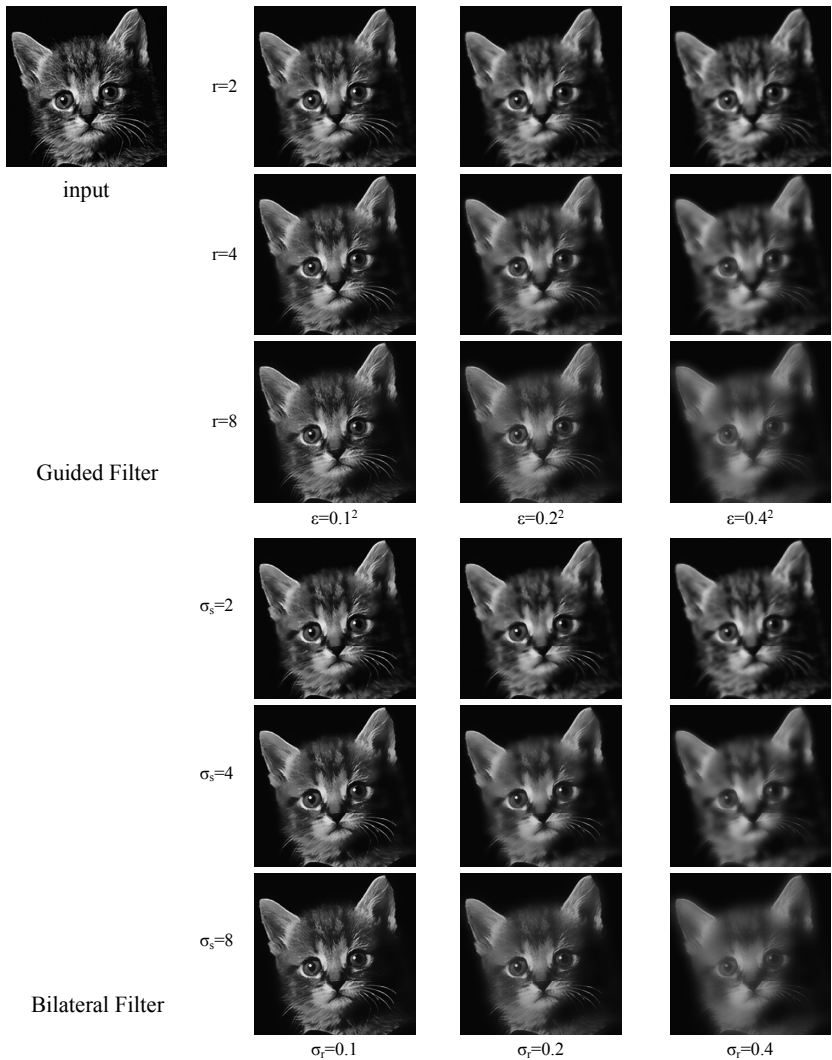


Fig. 1. The filtered images of a gray-scale input. In this example the guidance I is identical to the input p . The input image has intensity in $[0, 1]$. The input image is from [1].

Fig. 3 shows two examples of the kernel shapes in real images. In the top row are the kernels near a step edge. Like the bilateral kernel, the guided filter’s kernel assigns nearly zero weights to the pixels on the opposite side of the edge. In the bottom row are the kernels in a patch with small scale textures. Both filters average almost all the nearby pixels together and appear as low-pass filters.

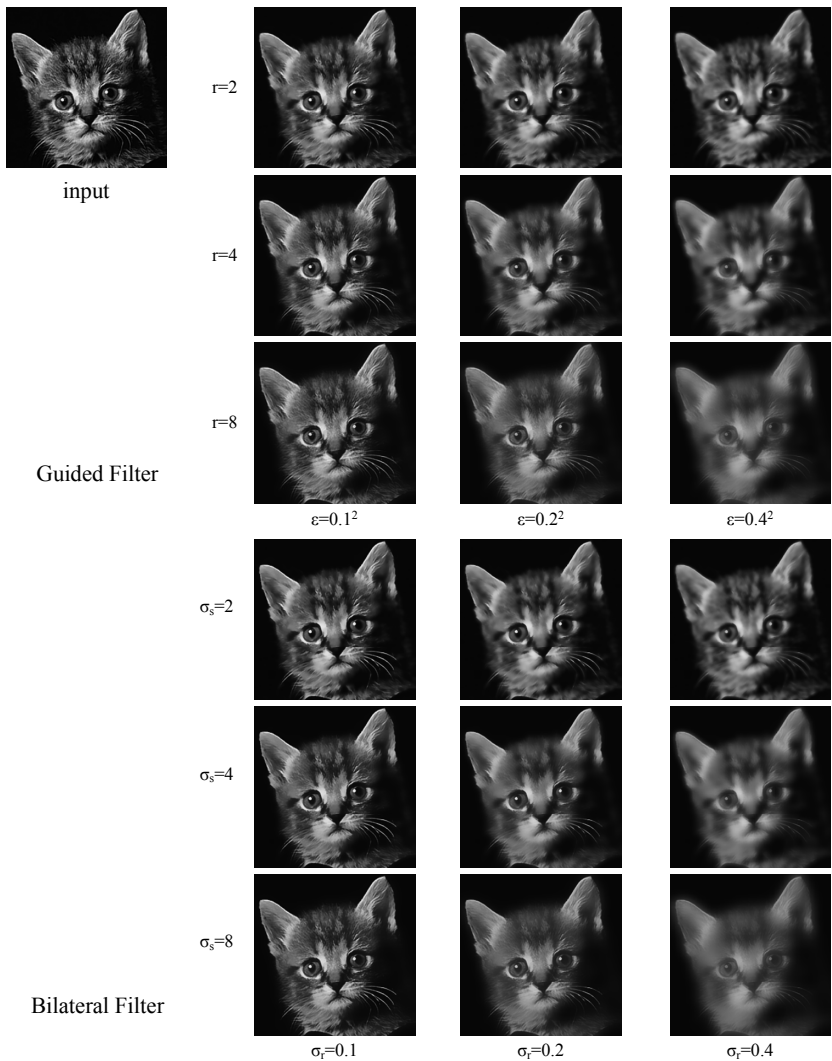


Fig. 1. 灰度输入图像的滤波结果。在此示例中，引导图 I 与输入图 p 相同。输入图像的强度范围为 $[0, 1]$ 。输入图像来自文献[1]。

图3展示了实际图像中两种核形状的例子。第一行是靠近阶梯边缘处的核，与双边滤波器类似，引导滤波器在边缘另一侧的像素上分配几乎为零的权重。第二行是在包含小尺度纹理补丁中的核，这两种滤波器都会平均化几乎所有附近的像素，并表现出低通滤波器的特性。

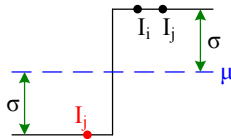


Fig. 2. A 1-D example of an ideal step edge. For a window that exactly center on the edge, the variables μ and σ are as indicated.

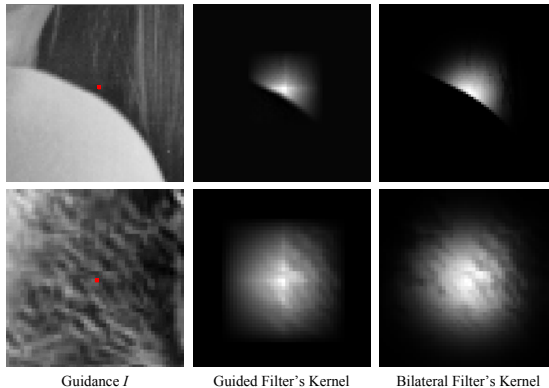


Fig. 3. Filter kernels. Top: a step edge (guided filter: $r = 7, \epsilon = 0.1^2$, bilateral filter: $\sigma_s = 7, \sigma_r = 0.1$). Bottom: a textured patch (guided filter: $r = 8, \epsilon = 0.2^2$, bilateral filter: $\sigma_s = 8, \sigma_r = 0.2$). The kernels are centered at the pixels denote by the red dots.

3.4 Gradient Preserving Filtering

Though the guided filter is an edge-preserving smoothing filter like the bilateral filter, it avoids the gradient reversal artifacts that may appear in detail enhancement and HDR compression. Fig. 4 shows a 1-D example of detail enhancement. Given the input signal (black), its edge-preserving smoothed output is used as a *base layer* (red). The difference between the input signal and the base layer is the *detail layer* (blue). It is magnified to boost the details. The enhanced signal (green) is the combination of the boosted detail layer and the base layer. An elaborate description of this method can be found in [12].

For the bilateral filter (Fig. 4 left), the base layer is not consistent with input signal at the edge pixels. This is because few pixels around them have similar colors, and the Gaussian weighted average has little statistical data and becomes unreliable. So the detail layer has great fluctuations, and the recombined signal has reversed gradients as shown in the figure. On the other hand, the guided filter (Fig. 4 right) better preserves the gradient information in I , because the gradient of the base layer is $\nabla q \approx \bar{a} \nabla I$ near the edge. The shape of the edge is well maintained in the recombined layer.

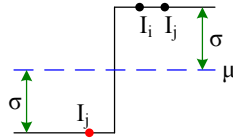


Fig. 2. 一维理想步阶边缘示例。对于恰好位于边缘中心的窗口，变量 μ 和 σ 如图所示。

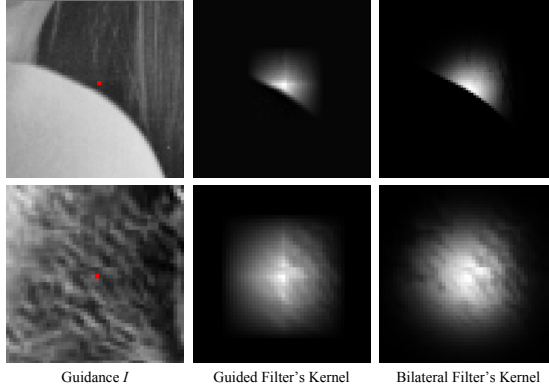


Fig. 3. 滤波核。上图：一个阶跃边缘（引导滤波器： $r = 7, \epsilon = 0.1^2$ ，双边滤波器： $\sigma_s = 7, \sigma_r = 0.1$ ）。下图：一个纹理块（引导滤波器： $r = 8, \epsilon = 0.2^2$ ，双边滤波器： $\sigma_s = 8, \sigma_r = 0.2$ ）。这些核以红色点表示的像素为中心。

3.4 Gradient Preserving Filtering

尽管导向滤波器是一种像双边滤波器那样的边缘保持平滑滤波器，但它避免了在细节增强和高动态范围压缩中可能出现的梯度反转伪影。图4展示了细节增强的一个一维示例。给定输入信号（黑色），其边缘保持平滑输出用作红色部分 *base layer* (red)。输入信号与基础层之间的差异为蓝色部分 *detail layer* (blue)，该差异被放大以增强细节。增强后的信号（绿色）是放大后的细节层和基础层的组合。有关此方法的详细描述可参见文献[12]。

对于双边滤波器（图4左），基础层在边缘像素处与输入信号不一致。这是因为在这些像素周围的少数几个像素颜色相似，高斯加权平均值缺乏足够的统计数据而变得不可靠。因此细节层波动较大，重组后的信号如图所示出现了反向梯度。另一方面，引导滤波器（图4右）更好地保留了 I 中的梯度信息，因为基础层在边缘附近的梯度为 $\nabla q \approx \bar{a} \nabla I$ ，重组后层次很好地保持了边缘的形状。

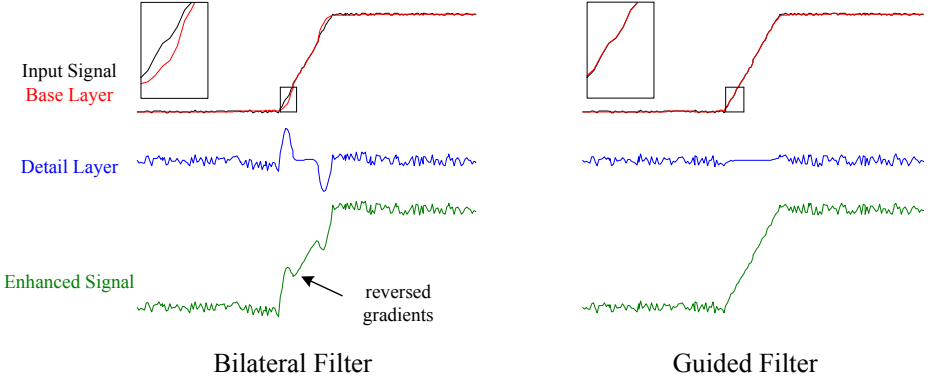


Fig. 4. 1-D illustration for detail enhancement. See the text for explanation.

3.5 Relation to the Matting Laplacian Matrix

The guided filter can not only be used as a smoothing operator. It is also closely related to the matting Laplacian matrix [2]. This casts new insights into the guided filter and inspires some new applications.

In a closed-form solution to matting [2], the matting Laplacian matrix is derived from a local linear model. Unlike the guided filter which computes the local optimal for each window, the closed-form solution seeks a global optimal. To solve for the unknown alpha matte, this method minimizes the following cost function:

$$E(\alpha) = (\alpha - \beta)^T \Lambda (\alpha - \beta) + \alpha^T L \alpha, \quad (10)$$

where α is the unknown alpha matte denoted in its matrix form, β is the constraint (e.g., a trimap), L is an $N \times N$ matting Laplacian matrix, and Λ is a diagonal matrix encoded with the weights of the constraints. The solution to this optimization problem is given by solving a linear system: $(L + \Lambda)\alpha = \Lambda\beta$.

The elements of the matting Laplacian matrix are given by:

$$L_{ij} = \sum_{k: (i,j) \in \omega_k} \left(\delta_{ij} - \frac{1}{|\omega|} \left(1 + \frac{(I_i - \mu_k)(I_j - \mu_k)}{\sigma_k^2 + \epsilon} \right) \right). \quad (11)$$

where δ_{ij} is the Kronecker delta. Comparing (11) with (9), we find that the elements of the matting Laplacian matrix can be directly given by the guided filter kernel weights:

$$L_{ij} = |\omega|(\delta_{ij} - W_{ij}), \quad (12)$$

Following the strategy in [24], we can further prove (see the supplementary materials) that the output of the guided filter is one Jacobi iteration in optimizing (10). If β is a reasonably good guess of the matte, we can run one Jacobi step and obtain an approximate solution to (10) by a guided filtering process: $\alpha_i \approx \sum_j W_{ij}(I) \beta_j$. In Section 4, we apply this property to image matting/feathering and haze removal.

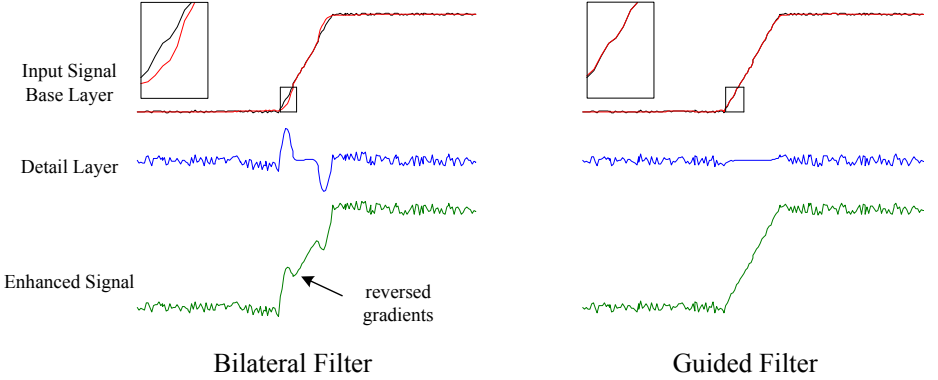


Fig. 4. 请提供需要翻译的英文意图文并观察中说明提供的信息中缺少具体的待翻译文本。

3.5 Relation to the Matting Laplacian Matrix

引导滤波器不仅可以用作平滑算子。它还与蒙版拉普拉斯矩阵[2]密切相关。这为引导滤波器提供了新的见解，并启发了一些新应用。

在求解蒙版问题的闭式解中[2]，蒙版拉普拉斯矩阵是从局部线性模型推导出来的。与引导滤波器逐窗口计算局部最优解不同，闭式解方法寻求全局最优解。为了求解未知的 α 蒙版，该方法最小化以下代价函数：

$$E(\alpha) = (\alpha - \beta)^T \Lambda (\alpha - \beta) + \alpha^T L \alpha, \quad (10)$$

其中， α 表示未知的 α 蒙版（以矩阵形式表示）， β 是约束条件（例如，三值图）， L 是一个 $N \times N$ 的抠图拉普拉斯矩阵，而 Λ 是一条对角线矩阵，其中编码了约束的权重。该优化问题的解可以通过求解以下线性系统得到： $(L + \Lambda)\alpha = \Lambda\beta$ 。

拉普拉斯矩阵的元素给定如下：

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$$L_{ij} = \sum_{k: (i,j) \in \omega_k} \left(\delta_{ij} - \frac{1}{|\omega|} \left(1 + \frac{(I_i - \mu_k)(I_j - \mu_k)}{\sigma_k^2 + \epsilon} \right) \right). \quad (11)$$

其中 δ_{ij} 是克罗内克 δ 。将 (11) 与 (9) 进行比较，我们发现基质拉普拉斯矩阵的元素可以直接由引导滤波核权重给出：

$$L_{ij} = |\omega| (\delta_{ij} - W_{ij}), \quad (12)$$

根据文献[24]中的策略，我们进一步证明（详见补充材料），引导滤波器的输出相当于优化方程(10)的一次Jacobi迭代。如果 β 是蒙版图的一个合理猜测，则可以通过一次Jacobi步骤并利用引导滤波过程获得方程(10)的大致解：

$\alpha_i \approx \sum_j W_{ij}(I)\beta_{j_0}$ 在第4节中，我们将这一性质应用于图像抠图/羽化以及雾霾去除。

3.6 O(N) Time Exact Algorithm

One more advantage of the guided filter over the bilateral filter is that it automatically has an $O(N)$ time exact algorithm. $O(N)$ time implies that the time complexity is independent of the window radius r , so we are free to use arbitrary kernel sizes in the applications.

The filtering process in (1) is a translation-variant convolution. Its computational complexity increases when the kernel becomes larger. Instead of directly performing the convolution, we compute the filter output from its definition (5)(6)(8). All the summations in these equations are box filters ($\sum_{i \in \omega_k} f_i$). We apply the $O(N)$ time Integral Image technique [28] to calculate the output of a box filter. So the guided filter can be computed in $O(N)$ time.

The $O(N)$ time algorithm can be easily extended to RGB color guidance images. Filtering using color guidance images is necessary when the edges or details are not discriminable in any single channel. To generalize to a color guidance image, we rewrite the local linear model (3) as:

$$q_i = \mathbf{a}_k^T \mathbf{I}_i + b_k, \forall i \in \omega_k. \quad (13)$$

Here $\mathbf{I}_i$ is a 3×1 color vector, $\mathbf{a}_k$ is a 3×1 coefficient vector, q_i and b_k are scalars. The guided filter for color guidance images becomes:

$$\mathbf{a}_k = (\Sigma_k + \epsilon \mathbf{U})^{-1} \left(\frac{1}{|\omega|} \sum_{i \in \omega_k} \mathbf{I}_i p_i - \mu_k \bar{p}_k \right) \quad (14)$$

$$b_k = \bar{p}_k - \mathbf{a}_k^T \mu_k \quad (15)$$

$$q_i = \bar{\mathbf{a}}_i^T \mathbf{I}_i + \bar{b}_i. \quad (16)$$

Here Σ_k is the 3×3 covariance matrix of $\mathbf{I}$ in ω_k , and $\mathbf{U}$ is a 3×3 identity matrix. The summations are still box filters and can be computed in $O(N)$ time.

We experiment the running time in a laptop with a 2.0Hz Intel Core 2 Duo CPU. For the gray-scale guided filter, it takes 80ms to process a 1-megapixel image. As a comparison, the $O(N)$ time bilateral filter in [15] requires 42ms using a histogram of 32 bins, and 85ms using 64 bins. Note that the guided filter algorithm is non-approximate and applicable for data of high bit-depth, while the $O(N)$ time bilateral filter may have noticeable quantization artifacts (see Fig. 5). The algorithm in [16] requires 1.2 seconds per megapixel using 8 bins (using the public code on the authors' website). For RGB guidance images, the guided filter takes about 0.3s to process a 1-megapixel image. The algorithm for high-dimensional bilateral filter in [16] takes about 10 seconds on average to process per 1-megapixel RGB image.

4 Applications and Experimental Results

In this section, we apply the guided filter to a great variety of computer vision and graphics applications.

3.6 O(N) Time Exact Algorithm

引导滤波相对于双边滤波的一个优势是它自动具有时间复杂度为 $O(N)$ 的精确算法。时间复杂度为 $O(N)$ 意味着计算时间与窗口半径 r 无关，因此在实际应用中我们可以自由选择任意大小的核函数。

滤波过程如公式(1)所示，是一种非平移不变的卷积。当核函数变大时，其计算复杂度会增加。我们不直接进行卷积运算，而是根据定义（公式5、6和8）来计算滤波器输出。这些方程中的所有求和操作都是盒式滤波器（ $\sum_{i \in \omega_k} f_i$ ）。我们应用 $O(N)$ 时间的积分图像技术[28]来计算盒式滤波器的输出，因此导向滤波器可以在 $O(N)$ 时间内完成计算。

时间复杂度为 $O(N)$ 的算法可以轻松扩展到RGB颜色引导图像。当任何单一通道中边缘或细节不可区分时，使用颜色引导图像进行滤波是必要的。为了将局部线性模型（公式3）推广到颜色引导图像，我们将其重写为：

$$q_i = \mathbf{a}_k^T \mathbf{I}_i + b_k, \forall i \in \omega_k. \quad (13)$$

这里 $\mathbf{I}_i$ 是一个 3×1 颜色向量， $\mathbf{a}_k$ 是一个 3×1 系数向量， q_i 和 b_k 是标量。用于颜色引导图像的导向滤波器变为：

$$\mathbf{a}_k = (\Sigma_k + \epsilon \mathbf{U})^{-1} \left(\frac{1}{|\omega|} \sum_{i \in \omega_k} \mathbf{I}_i p_i - \mu_k \bar{p}_k \right) \quad (14)$$

$$b_k = \bar{p}_k - \mathbf{a}_k^T \mu_k \quad (15)$$

$$q_i = \bar{\mathbf{a}}_i^T \mathbf{I}_i + \bar{b}_i. \quad (16)$$

这里 Σ_k 是 $\mathbf{I}$ 在 ω_k 中的 3×3 协方差矩阵， $\mathbf{U}$ 是一个 3×3 的单位矩阵。求和运算仍然是盒式滤波器，可以在 $O(N)$ 时间内完成计算。

我们在配备2.0GHz Intel Core 2 Duo CPU的笔记本电脑上测试了运行时间。对于灰度引导滤波器，处理一张100万像素的图像需要80毫秒。相比之下，文献[15]中的 $O(N)$ 时间复杂度双边滤波器使用32个直方图分箱时耗时42毫秒，而使用64个分箱时则需85毫秒。需要注意的是，引导滤波算法是非近似且适用于高比特深度的数据，而 $O(N)$ 时间复杂度的双边滤波器可能会有明显的量化伪影（见图5）。文献[16]中的方法每处理一张百万像素图像需要大约1.2秒（使用作者网站上的公开代码，并采用8个分箱）。对于RGB引导图像，引导滤波器处理一张100万像素的图像大约需要0.3秒。文献[16]中提出的高维双边滤波算法平均每处理一张百万像素的RGB图像耗时约10秒。

4 Applications and Experimental Results

在本节中，我们将引导滤波器应用于各种计算机视觉和图形学应用。



Fig. 5. Quantization artifacts of $O(N)$ time bilateral filter. (a) Input HDR image (32bit float, displayed by linear scaling). (b) Compressed image using the $O(N)$ bilateral filter in [15] (64 bins). (c) Compressed image using the guided filter. This figure is best viewed in the electronic version of this paper.

Detail Enhancement and HDR Compression. The method for detail enhancement is described in Section 3.4. For HDR compression, we compress the base layer instead of magnifying the detail layer. Fig. 6 shows an example for detail enhancement, and Fig. 7 shows an example for HDR Compression. The results using the bilateral filter are also provided. As shown in the zoom-in patches, the bilateral filter leads to gradient reversal artifacts.



Fig. 6. Detail enhancement. The parameters are $r = 16$, $\epsilon = 0.1^2$ for the guided filter, and $\sigma_s = 16$, $\sigma_r = 0.1$ for the bilateral filter. The detail layer is boosted $\times 5$.

Flash/No-flash Denoising. In [11] it is proposed to denoise a no-flash image under the guidance of its flash version. Fig. 8 shows a comparison of using the joint bilateral filter and the guided filter. The gradient reversal artifacts are noticeable near some edges in the joint bilateral filter result.

Matting/Guided Feathering. We apply the guided filter as *guided feathering*: a binary mask is refined to appear an alpha matte near the object boundaries



Fig. 5. 量化效应的 $O(N)$ 时间双边滤波器。(a) 输入HDR图像（32位浮点，通过线性缩放显示）。(b) 使用文献[15]中的 $O(N)$ 双边滤波器压缩后的图像（64个区间）。(c) 使用引导滤波器压缩后的图像。请参阅本文的电子版以获得最佳视觉效果。

Detail Enhancement and HDR Compression. 详细增强的方法在第3.4节中进行了描述。对于HDR压缩，我们压缩基础层而不是放大细节层。图6展示了详细增强的一个示例，而图7则展示了HDR压缩的示例。双边滤波器的结果也一并提供。如放大后的图像块所示，双边滤波器会导致梯度反转伪影。



Fig. 6. 细节增强。引导滤波器的参数为 $r = 16$, $\epsilon = 0.1^2$ ，双边滤波器的参数为 $\sigma_s = 16$, $\sigma_r = 0.1$ 。细节层被增强倍数为 $\times 5$ 。

Flash/No-flash Denoising. 在文献[11]中，提出了一种在闪光版本的指导下对无闪光图像进行去噪的方法。图8展示了联合双边滤波器和引导滤波器的对比。在联合双边滤波器的结果中，某些边缘附近可以明显观察到梯度反转伪影。

Matting/Guided Feathering. 我们将引导滤波器应用于 *guided feathering*: 一个二值掩码被细化，使其在对象边界附近呈现为alpha蒙版。

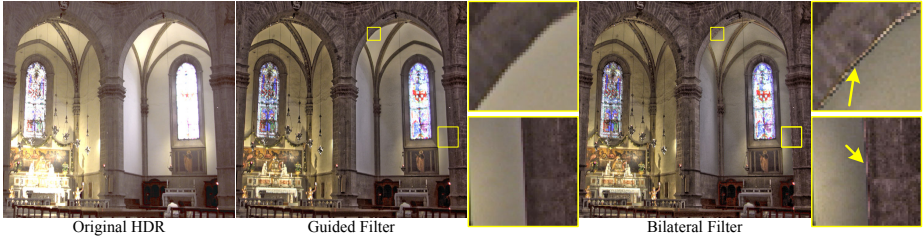


Fig. 7. HDR compression. The parameters are $r = 15$, $\epsilon = 0.12^2$ for the guided filter, and $\sigma_s = 15$, $\sigma_r = 0.12$ for the bilateral filter.

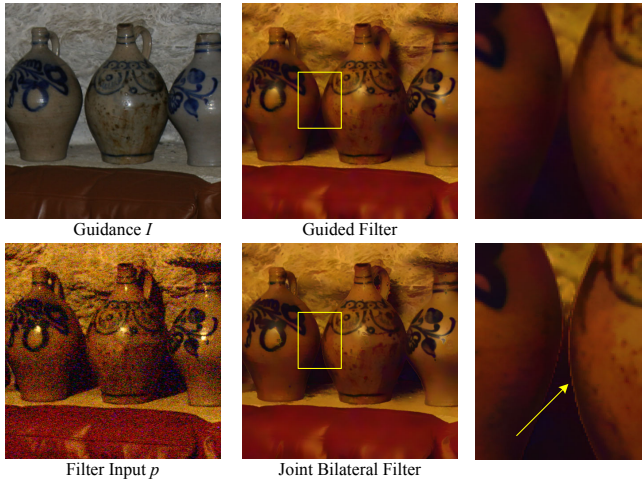


Fig. 8. Flash/no-flash denoising. The parameters are $r = 8$, $\epsilon = 0.2^2$ for the guided filter, and $\sigma_s = 8$, $\sigma_r = 0.2$ for the joint bilateral filter.

(Fig. 9). The binary mask can be obtained from graph-cut or other segmentation methods, and is used as the filter input p . The guidance $\mathbf{I}$ is the color image. A similar function “Refine Edge” can be found in the commercial software Adobe Photoshop CS4. We can also compute an accurate matte using the closed-form solution [2]. In Fig. 9 we compare our results with the Photoshop Refine Edge and the closed-form solution. Our result is visually comparable with the closed-form solution in this short hair case. Both our method and Photoshop provide fast feedback ($<1s$) for this 6-mega-pixel image, while the closed-form solution takes about two minutes to solve a huge linear system.

Single Image Haze Removal. In [9] a haze transmission map is roughly estimated using a dark channel prior, and is refined by solving the matting Laplacian matrix. On the contrary, we simply filter the raw transmission map under the guidance of the hazy image. The results are visually similar (Fig. 10). The guid-

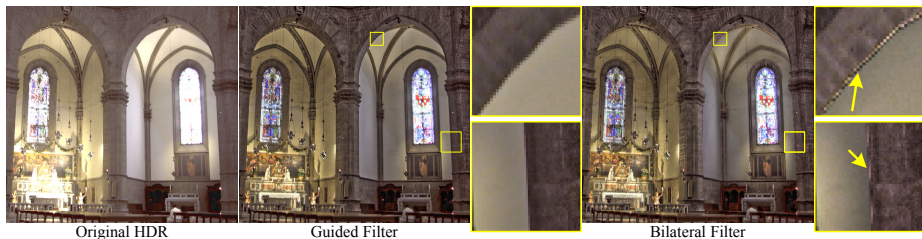


Fig. 7. HDR 压缩。参数为：引导滤波器的 $r = 15$, $\epsilon = 0.12^2$ ；双边滤波器的 $\sigma_s = 15$, $\sigma_r = 0.12$ 。

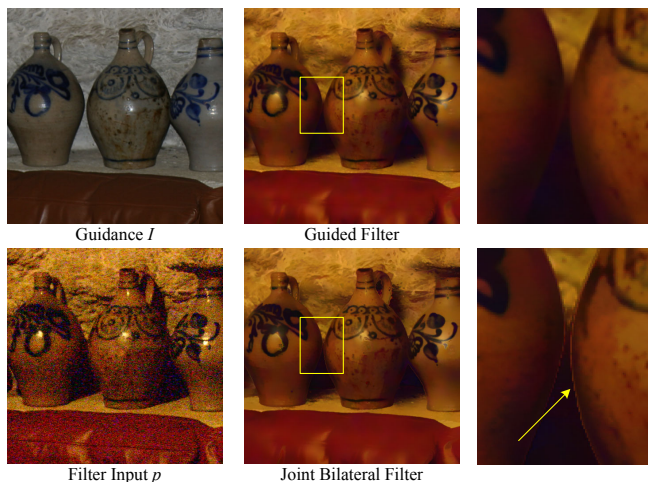


Fig. 8. 闪存/无闪存去噪。引导滤波器的参数为 $r = 8$, $\epsilon = 0.2^2$ ；联合双边滤波器的参数为 $\sigma_s = 8$, $\sigma_r = 0.2$ 。

(图9) 二值掩码可以通过图割或其他分割方法获得，并用作滤镜输入 p 。引导信息 I 是彩色图像。在商业软件 Adobe Photoshop CS4 中也可以找到类似的功能“锐化边缘”。我们还可以使用封闭形式解 [2] 计算准确的蒙版。在图 9 中，我们将我们的结果与 Photoshop 锐化边缘功能和封闭形式解进行了比较。在短发案例中，我们的结果在视觉上与封闭形式解相当。对于这张 600 万像素的图像，无论是我们的方法还是 Photoshop 都可以提供快速反馈 (<1 秒)，而封闭形式解则需要大约两分钟来求解一个巨大的线性系统。

Single Image Haze Removal. 在文献[9]中，雾传输图通过暗通道先验粗略估计，并通过求解蒙板拉普拉斯矩阵进行优化。相反，我们在雾霾图像的指导下直接过滤原始传输图。结果视觉上相似（见图10）。指导原则为：

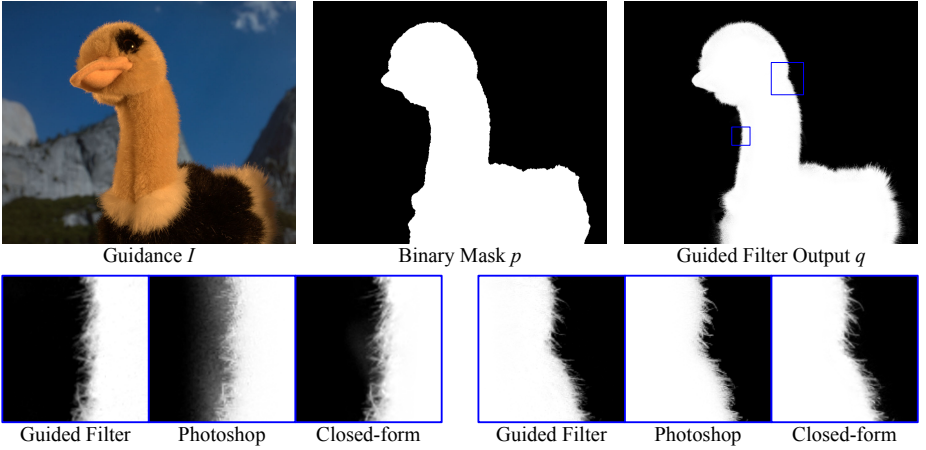


Fig. 9. Guided Feathering. A binary mask p is filtered under the guidance of I . In the zoom-in patches, we compare with the Photoshop Refine Edge function and the closed-form matting. For closed-form matting, we erode and dilate the mask to obtain a trimap. The parameters are $r = 60$, $\epsilon = 10^{-6}$ for the guided filter.

ed filter takes about 0.1s to process this 600×400 color image, but the running time is over 10 seconds as reported in [9] .

Joint Upsampling. Joint upsampling [21] is to upsample an image under the guidance of another image. Taking the application of colorization [7] as an example. A gray-scale luminance image is colorized through an optimization process. To reduce the running time, the chrominance channels are solved at a coarse resolution and upsampled under the guidance of the full resolution luminance image by the joint bilateral filter [21]. This upsampling process can also be performed by the guided filter. The result is visually comparable (Fig. 11).

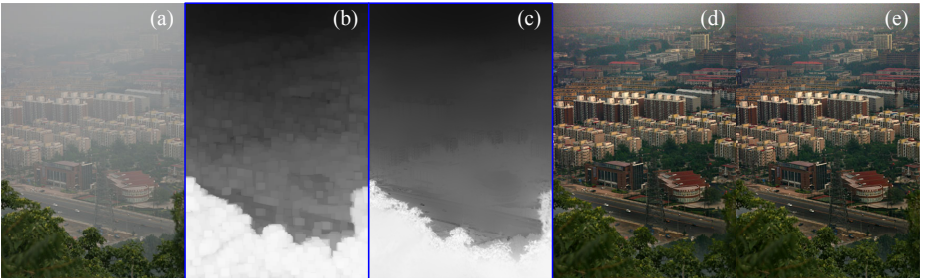


Fig. 10. Haze Removal. (a) Hazy image. (b) Raw transmission map [9]. (c) The raw transmission map is refined by the guided filter ($r = 20$, $\epsilon = 10^{-3}$). (e) Recovered image using (c). (d) The result in [9].

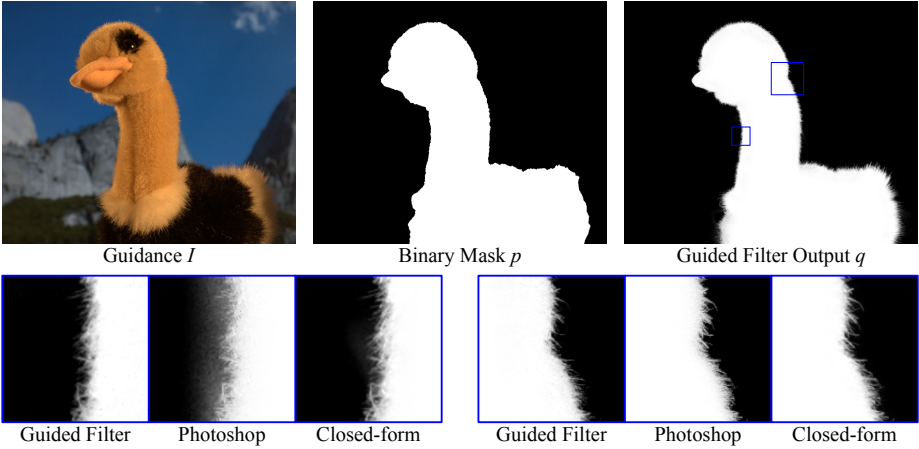


Fig. 9. 引导式羽化。在二值掩码 p 的基础上，根据 I 进行滤波处理。在放大区域中，我们与 Photoshop 的“细化边缘”功能以及闭式抠图方法进行比较。对于闭式抠图，我们通过腐蚀和膨胀操作获取三值图（trimap）。引导滤波的参数设置为 $r = 60, \epsilon = 10^{-6}$ 。

ed滤波器处理这张600×400彩色图像大约需要0.1秒，但据文献[9]报道的运行时间超过10秒。

Joint Upsampling. 联合上采样[21]是在另一张图像的指导下对图像进行上采样的过程。以色彩化应用[7]为例，通过优化过程将灰度亮度图转换为彩色图。为了减少运行时间，在较低分辨率下求解色度通道，并在全分辨率亮度图像的指导下使用联合双边滤波器[21]对其进行上采样。此上采样过程也可以由引导滤波器完成，结果视觉效果相当（见图11）。

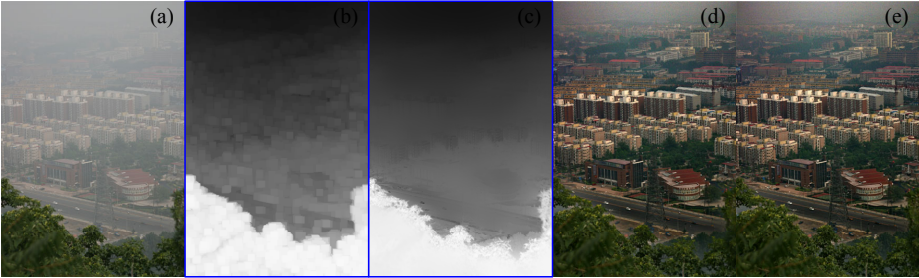


Fig. 10. 雾霾去除。(a) 雾霾图像。(b) 原始透传图[9]。(c) 使用引导滤波器对原始透传图进行优化 ($r = 20, \epsilon = 10^{-3}$)。 (e) 利用(c)恢复的图像。(d) [9]中的结果。

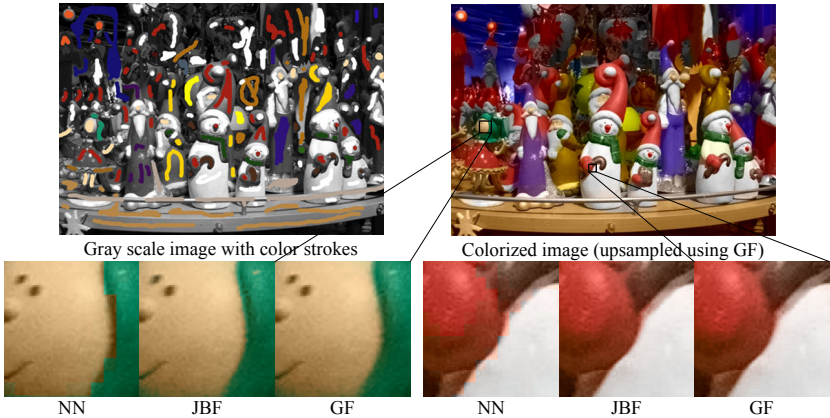


Fig. 11. Joint Upsampling for Colorization. The upsampling methods includes: nearest-neighbor (NN), joint bilateral filter (JBF), and guided filter (GF).

5 Discussion and Conclusion

In this paper, we have presented a novel filter which is widely applicable in computer vision and graphics. Different from the recent trend towards accelerating the bilateral filter [14–17], we define a new type of filter that shares the nice property of edge-preserving smoothing but can be computed efficiently and exactly. Our filter is more generic and can handle some applications beyond the concept of “smoothing”. Since the local linear model (3) can be regarded as a simple case of learning, other advanced models/features might be applied to obtain new filters.

As a locally based operator, the guided filter is not directly applicable for sparse inputs like strokes. It also shares a common limitation of other explicit filter - it may have halos near some edges. In fact, it is ambiguous for a low-level and local operator to determine which edge should be smoothed and which should be preserved. Unsuitably smoothing an edge will result in halos near it. However, we believe that the simplicity and efficiency of the guided filter still make it beneficial in many situations.

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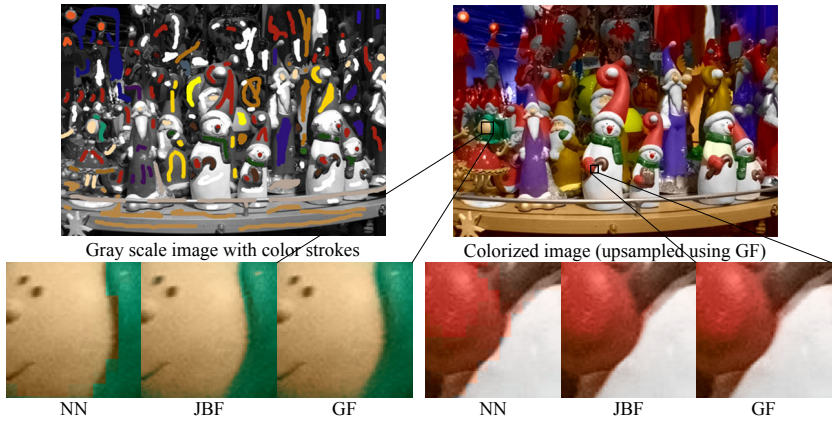


Fig. 11. 色彩化联合上采样方法包括：最近邻（NN）、联合双边滤波器（JBF）和引导滤波器（GF）。

5 Discussion and Conclusion

在本文中，我们提出了一种新颖的滤波器，该滤波器广泛应用于计算机视觉和图形学领域。与近期加速双边滤波器的趋势[14–17]不同，我们定义了一种新型滤波器，它具有边缘保持平滑的优点，并且可以高效准确地计算。我们的滤波器更为通用，能够处理超出“平滑”概念的应用场景。由于局部线性模型（3）可被视为一种简单的学习案例，因此其他高级模型/特征也可以应用以获得新的滤波器。

作为局部操作符，引导滤波器不直接适用于稀疏输入（如笔画）。它与其他显式滤波器一样存在一个共同的局限性——在某些边缘附近可能会产生晕轮效应。实际上，对于低级和局部的操作符来说，确定哪些边缘需要平滑处理而哪些应该保留是模糊不清的。如果错误地对某个边缘进行平滑处理，则会在该边缘周围产生晕轮效应。然而，我们认为引导滤波器的简洁性和高效性仍然使其在许多情况下具有益处。

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